Reproducible Evaluation Protocol

1. Record date, tested Git commit SHA, Python version, provider/model, retrieval top_k and threshold.

2. Run python -m pytest -q at repo root; store dated output and failures.

3. Ingest braking-system test PDF (../data/samples/AUTOSAR_HLD_Sample_BrakingSystem.pdf).

4. Evaluate golden dataset (../data/golden_dataset.json) through PipelineEvaluator in core/evaluation/evaluator.py (../core/evaluation/evaluator.py), if the current environment supports it.

5. Record retrieved chunk IDs and expected page match for answerable items.

6. Inspect whether each generated claim is actually entailed by cited text; valid IDs alone are insufficient.

7. For unanswerable questions, demand explicit insufficient-evidence language, without invented specifications.

8. Distinguish MockLLM vs Gemini evaluations; measure latency, refusal correctness, retrieval success, invalid citations and human-reviewed answer quality.

9. Save screenshots/logs only after redacting any API key or sensitive PDF text.

Implemented automated evaluator scores: retrieval success rate, groundedness flag rate, citation ID validity rate, negative-grounding accuracy and expected-keyword overlap. None alone is an automotive-compliance or calibrated factual-accuracy measure. A previous historic statement was “70 passed, 1 skipped”; rerun locally before claiming current performance.

Current Evaluation Status — Historical vs Verified

Check	Status / interpretation
Source code	The GitHub repository contains backend, frontend, core, storage, tests and golden dataset
Historical pytest status	Student previously reported 70 passed / 1 skipped after Phase 7; not freshly rerun by this documentation commit
Golden data	Ten-question synthetic braking dataset is checked in
Door Control PDF manual test	User reported successful upload and 16 Chroma chunks; the PDF itself is not currently in this repo
Positive MockLLM question	Relevant chunk and reference ID returned, but answer wording was generic
Negative MockLLM MCU question	Generic cited answer appeared, rather than the desired abstention: does not demonstrate refusal success
Gemini behavior	Has not been independently measured in this documentation update
Citation verifier	Checks referenced chunk IDs, not whether the cited text actually supports every claim

Do not publish fabricated accuracy numbers. Append fresh time-stamped results to RESULTS_TO_FILL.csv (RESULTS_TO_FILL.csv).

Golden Dataset Guide

Source of truth: data/golden_dataset.json (../data/golden_dataset.json), ten questions about a synthetic braking-system HLD.

• Questions q1–q7: answerable component responsibilities, P_BrakePressure, R_PedalPosition, If_BrakePressure, related ports, sensor/actuator descriptions.

• Questions q8–q10: deliberately unanswerable unrelated questions about satellite encryption, jet-engine afterburner ratio and battery cells.

The JSON includes expected evidence page, chunk IDs, answer summaries and keyword expectations. Do not conflate these with a separate Door Control demo document. Expected data is a test specification, not an observed performance result.